

Supplementary Information

An open-source, 3D-printed auger powder doser designed with generative artificial intelligence

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S1 Bill of materials

Table S1 lists the parts for one channel, plus the items a platform needs only once whatever its number of channels. Prices are single-unit US dollars as of September 2026 and should be checked at order time; vendor links are kept in hardware/ in the repository (<https://github.com/vertical-cloud-lab/powder-doser>).

Table S1: Bill of materials. Printed parts are excluded (about 260 g of PLA per channel, under \$10), as are the custom circuit board, which can be replaced by a breadboard, and the analytical balance (A&D HR-100A, 0.1 mg readability), which most laboratories already own.

Part	Qty	Price (USD)	Source
<i>Per channel</i>			
NEMA-11 bipolar stepper, 0.67 A, 10 N-cm (11HS18-0674S)	1	14	StepperOnline
Pololu Tic T500 stepper controller	1	33	Pololu #3135
Pololu 33 V shunt regulator (clamps stepper back-voltage)	1	15	Pololu #3776
12 V push-pull solenoid, JF-0530B type (tapping)	1	7.50	Adafruit #412
DRV8871 H-bridge breakout (solenoid driver)	1	7.50	Adafruit #3190
ERM coin vibration motor, 10 mm	1	1.95	Adafruit #1201
DRV2605L haptic driver breakout (I ² C)	1	7.95	Adafruit #2305
MG996R-class metal-gear servo (tilt)	2	24	generic
Fasteners (M3), bulk capacitors	—	5	generic
<i>Shared (one per platform)</i>			
Raspberry Pi Pico W	1	6	Raspberry Pi
Waveshare Pico-2CH-RS232 board (balance interface)	1	9	Waveshare 19979
Mean Well GST60A12-P1J 12 V/5 A supply, cord and pigtail	1	26	Digi-Key
Pololu D24V22F5 5 V buck regulator	1	19	Pololu #2858
Headers and wiring	—	10	generic
Total, single-channel platform		≈185	

S2 Construction guide

An illustrated build guide is kept in the repository alongside the parametric CAD sources and STL files for every part. In outline:

1. **Print** the auger, the 20-tooth pinion, the bracket and collars, the tap collar (two clamp halves), the mounting plate, the hinged baseplate and the servo gear sector in PLA (0.2 mm layers, four perimeters, 25% infill, no supports). Print the auger upright.
2. **Auger drive:** seat the auger in its printed collars on the mounting plate, bolt the NEMA-11 stepper with its pinion to the plate, and check that the pinion meshes with the auger’s 44-tooth gear band (2.2:1).

3. **Tap collar:** clamp the two collar halves around the auger tube, mount the solenoid on the collar flange (2×M3) so that its plunger strikes the anvil pad, and stick the vibration motor to the opposite pad.
4. **Tilt stage:** join the mounting plate to the baseplate through the side hinges (printed pins), mesh the two servos with the plate's gear (2:1), and check that the outlet lies on the hinge axis.
5. **Electronics:** wire the Pico W, Tic T500 (serial), DRV8871, DRV2605L (I²C), the RS-232 board (to the HR-100A balance) and the servos as shown in the schematic in `hardware/test-module/`, then flash the MicroPython firmware.
6. **Checkout:** run the firmware self-test (rotate, tap, vibrate, tilt, tare and read), then the test protocols listed in the main text.

S3 Auger and outlet geometry

Fig. S1 shows a cross-section of the geared auger and the four printed outlet variants. The tests in the main text used the Type 4 outlet, with a 3 mm exit hole.

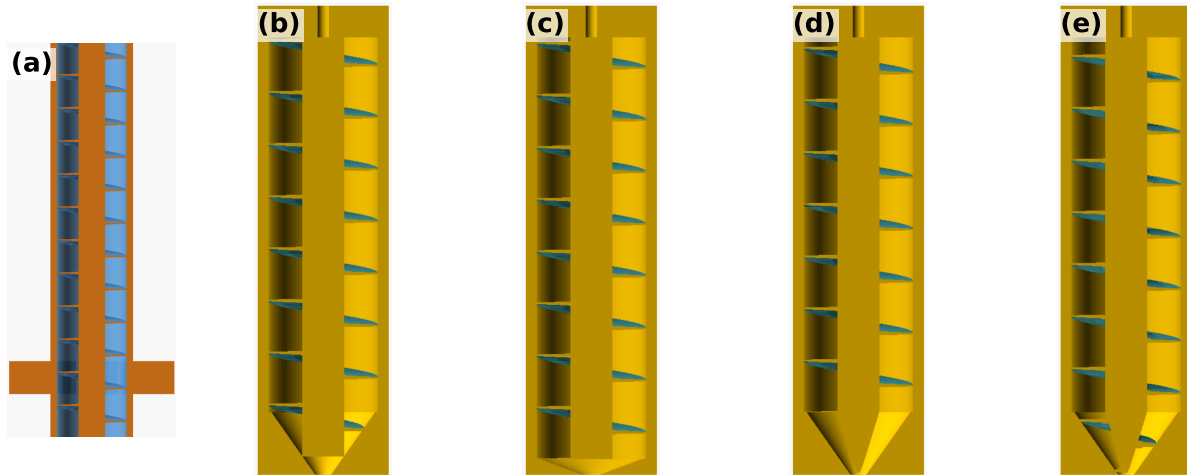


Figure S1: (a) Cross-section of the geared auger: the 8 mm central core, the single-start helical flight (10 mm pitch, 2 mm thick) fused to the 21 mm bore of the tube, and the external gear band. (b–e) Cross-sections of the four outlet variants (Types 1–4), which differ in exit constriction and chamfer. All are parametric OpenSCAD models (`cad/auger-geared/` in the repository) written by AI coding agents under human review; no conventional CAD software was used to design them. The auger used in the tests is the later Fusion 360 recreation of this part, whose gear band has 44 teeth rather than the 48 of the AI-modelled band.

S4 Bench noise

Protocol A takes eight balance readings with nothing moving at the start of every run. Their spread sets the smallest change that run can resolve (Fig. S2). It was 0.0 mg on every run up to 12 August 2026 and rose to tens of milligrams from 20 August. Runs are compared against their own noise floor throughout the main text.

S5 Closed-loop dose record

Table S2 summarizes every valid closed-loop dose by powder and target mass. A dose is valid when its run passed quality screening, the dose ended in a defined controller state, and the auger or solenoid actually moved; doses excluded for any other reason (for example, a misaligned outlet, an unreadable balance or powder that caked in the tube) are listed with the reason in the per-dose file.

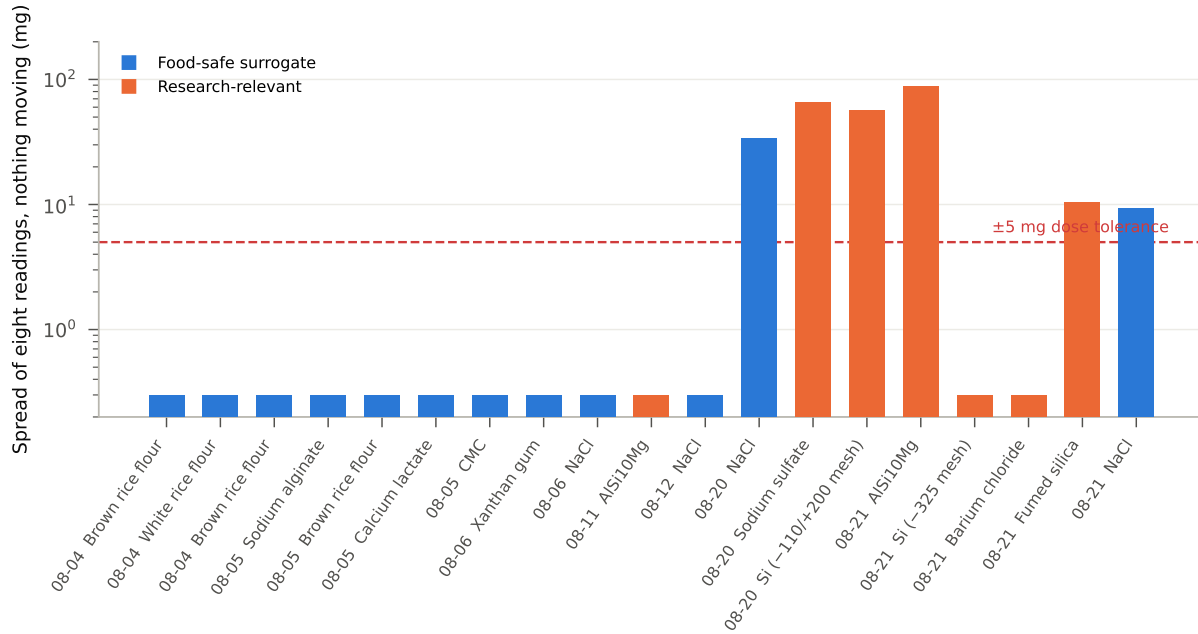


Figure S2: Balance noise floor for every first-round run: the spread (maximum minus minimum) of eight readings taken with nothing moving (protocol A), in date order. Runs whose eight readings were identical are drawn at the axis floor. The dashed line is the controller's ± 5 mg stopping band.

Table S2: Closed-loop dose summary for every valid dose (protocol G at 1 g; protocol H at 50 and 200 mg). Error is delivered minus requested mass. *Within limit*: doses inside the acceptance limit ($\pm 10\%$ at 50 mg, $\pm 5\%$ at 200 mg and 1 g). *Firmware ok*: doses the controller ended inside its own ± 5 mg stopping band. Round 1: August 2026, balance read on its stability flag; round 2: September 2026, bracketed readings. Per-dose records, including invalid doses and the reason each was excluded, are in `paper/figures/data/doses_all.csv`.

Powder	Target	Round	<i>n</i>	Mean error (mg)	SD (mg)	Within limit	Firmware ok	Median time (s)
AlSi10Mg	50 mg	2	3	+21.5	26.7	1/3	1/3	15
Si (-110/+200 mesh)	50 mg	2	2	+1.1	4.8	2/2	2/2	18
Sodium sulfate	50 mg	2	3	+28.9	15.7	0/3	0/3	10
Calcium lactate	50 mg	2	2	+9.8	15.8	1/2	1/2	30
Xanthan gum	50 mg	2	3	+17.8	36.4	2/3	2/3	37
NaCl	50 mg	2	3	+4.2	4.8	2/3	2/3	17
CMC	50 mg	2	3	-2.2	2.4	3/3	3/3	211
White rice flour	50 mg	2	3	-4.7	0.6	3/3	3/3	136
Sodium alginate	50 mg	2	3	-19.6	8.3	0/3	0/3	561
Si (-325 mesh)	50 mg	2	3	-50.0	0.1	0/3	0/3	31
AlSi10Mg	200 mg	2	3	-0.3	2.7	3/3	3/3	214
Si (-110/+200 mesh)	200 mg	2	3	+9.8	19.8	2/3	2/3	34
Sodium sulfate	200 mg	2	3	+10.8	17.7	2/3	1/3	30
Calcium lactate	200 mg	2	3	-16.5	16.6	2/3	1/3	464
Xanthan gum	200 mg	2	3	+2.9	0.6	3/3	3/3	89
NaCl	200 mg	2	3	+2.8	9.2	2/3	2/3	53
CMC	200 mg	2	3	-22.6	23.4	1/3	1/3	903
White rice flour	200 mg	2	3	-4.9	0.1	3/3	3/3	766
Sodium alginate	200 mg	2	3	-101.5	73.3	0/3	0/3	903
Si (-325 mesh)	200 mg	2	3	-199.7	0.5	0/3	0/3	31

Powder	Target	Round	<i>n</i>	Mean error (mg)	SD (mg)	Within limit	Firmware ok	Median time (s)
AlSi10Mg	1 g	2	3	+1.9	6.6	3/3	2/3	65
Si (−110/+200 mesh)	1 g	2	3	+0.2	2.8	3/3	3/3	128
Sodium sulfate	1 g	2	3	−4.2	1.1	3/3	3/3	86
Calcium lactate	1 g	1	3	−26.5	9.9	3/3	0/3	263
Xanthan gum	1 g	1	3	−32.9	9.6	3/3	0/3	193
NaCl	1 g	1	6	−1.9	5.3	6/6	5/6	186
CMC	1 g	1	3	−43.2	5.4	3/3	0/3	668
White rice flour	1 g	1	3	−137.9	23.5	0/3	0/3	822
Sodium alginate	1 g	1	3	−291.5	18.2	0/3	0/3	840
Brown rice flour	1 g	1	3	−999.1	0.9	0/3	0/3	7
Si (−325 mesh)	1 g	1,2	6	−999.5	0.7	0/6	0/6	8

S6 Run inventory

Table S3 lists every run recorded in the two test rounds, including runs that were aborted, postponed or excluded, so that each number in the main text can be traced to the run it came from.

Table S3: Every recorded run in both test rounds, in date order. *Protocols* lists the test protocols the run requested (Table 3 of the main text; protocol F never produced data); *Doses* is the number of closed-loop doses attempted. Locations: an open laboratory bench (to 6 August 2026), a shared fume hood (11 August to 3 September) and a second fume hood (from 3 September). Runs that failed screening are listed here and in `paper/figures/data/runs_all.csv` with the reason; the Experimental section describes how runs were chosen for each figure.

Start (UTC)	Round	Powder	Protocols	Location	Doses	Screening
2026-08-04 20:43	1	Brown rice flour	ABCDEFGF	open bench	3	excluded: nothing delivered, auger suspected
2026-08-04 21:14	1	White rice flour	ABCDEFGF	open bench	3	passed
2026-08-04 22:49	1	Brown rice flour	ABCDEFGF	open bench	3	excluded: nothing delivered, auger suspected
2026-08-05 14:57	1	Sodium alginate	ABCDEFGF	open bench	3	passed
2026-08-05 18:53	1	Brown rice flour	ABCDEFGF	open bench	3	passed (slow delivery)
2026-08-05 20:00	1	Calcium lactate	ABCDEFGF	open bench	3	passed
2026-08-05 21:12	1	—	—	open bench	0	aborted before testing
2026-08-05 21:52	1	CMC	ABCDEFGF	open bench	3	passed
2026-08-06 14:02	1	Xanthan gum	ABCDEFGF	open bench	3	passed
2026-08-06 14:51	1	NaCl	ABCDEFGF	open bench	3	passed
2026-08-11 18:30	1	AlSi10Mg	ABCDEFGF	fume hood 1	3	excluded: tilt servo failed
2026-08-12 21:51	1	NaCl	ABCDEFGF	fume hood 1	3	passed
2026-08-20 17:56	1	NaCl	ABCDE	fume hood 1	0	excluded: deliberate bench stress test
2026-08-20 19:22	1	Sodium sulfate	ABCDE	fume hood 1	0	passed
2026-08-20 21:48	1	Si (−110/+200 mesh)	ABCDE	fume hood 1	0	passed
2026-08-21 15:13	1	AlSi10Mg	ABCDE	fume hood 1	0	passed
2026-08-21 16:18	1	Si (−325 mesh)	ABCDEG	fume hood 1	3	passed (slow delivery)
2026-08-21 17:07	1	Barium chloride	ABCDE	fume hood 1	0	passed
2026-08-21 18:27	1	Fumed silica	ABCDE	fume hood 1	0	excluded: nothing delivered, outlet not checked
2026-08-21 19:20	1	NaCl	ABCDE	fume hood 1	0	passed
2026-09-03 16:35	2	NaCl	H	fume hood 1	6	excluded: balance could not be read
2026-09-03 17:04	2	NaCl	H	fume hood 1	6	excluded: balance could not be read
2026-09-03 21:40	2	NaCl	H	fume hood 2	6	passed
2026-09-04 00:18	2	NaCl	G	fume hood 2	1	demonstration only
2026-09-04 19:00	2	White rice flour	H	fume hood 2	6	passed
2026-09-04 20:37	2	Xanthan gum	H	fume hood 2	6	passed
2026-09-08 17:27	2	CMC	H	fume hood 2	6	excluded: outlet misaligned

Start (UTC)	Round	Powder	Protocols	Location	Doses	Screening
2026-09-08 19:42	2	CMC	H	fume hood 2	6	passed
2026-09-09 18:53	2	Sodium alginate	H	fume hood 2	6	passed
2026-09-09 22:52	2	Calcium lactate	H	fume hood 2	5	passed
2026-09-10 13:39	2	Sodium sulfate	GH	fume hood 2	9	passed
2026-09-10 17:08	2	Barium chloride	GH	fume hood 2	9	excluded: powder caked in the tube
2026-09-10 19:44	2	AlSi10Mg	GH	fume hood 2	9	passed
2026-09-10 21:29	2	Si (−325 mesh)	GH	fume hood 2	9	passed (slow delivery)
2026-09-10 23:19	2	—	—	fume hood 2	0	postponed: failed the stability check
2026-09-15 02:55	2	Si (−110/+200 mesh)	GH	fume hood 2	9	passed
2026-09-15 16:42	2	NaCl	G	fume hood 2	1	demonstration only

S7 AI usage log

In line with *Digital Discovery*’s guidance on the use of LLMs, all AI interactions are preserved publicly in the repository:

- **CAD and code generation:** every coding-agent session (GitHub Copilot coding agent and Claude-family models, 2026) is recorded as a pull request containing the prompt (issue text and review comments), the generated CAD code, rendered views and the human review thread.
- **Design reviews and literature queries:** Edison Scientific query scripts, task identifiers, raw answers and extracted references are archived with the corresponding branches; the consolidated bibliography, with the verification status of each entry, is `paper/references_validation.md`.
- **Text-to-CAD evaluations:** CADSmith run artifacts and the Zoo Design Studio (Zookeeper agent) session transcripts and notes are preserved on their branches.
- **Analysis and writing:** the data-analysis scripts behind every figure, and the AI-drafted revisions of this manuscript, are in `paper/` with their review threads.
- **Qualitative observations:** the team’s running journal of AI behaviour, good and bad, is discussion #39 in the repository.

Fig. S3 shows how agent use was spread over the project. Team members sent 754 requests to the two coding agents between April and September 2026, and the agents authored 789 of the 877 commits (90%). The counts cover all work in the repository (CAD, firmware, data analysis and writing), not only the design of the parts. The team moved from the GitHub Copilot coding agent to Claude at the start of July 2026.

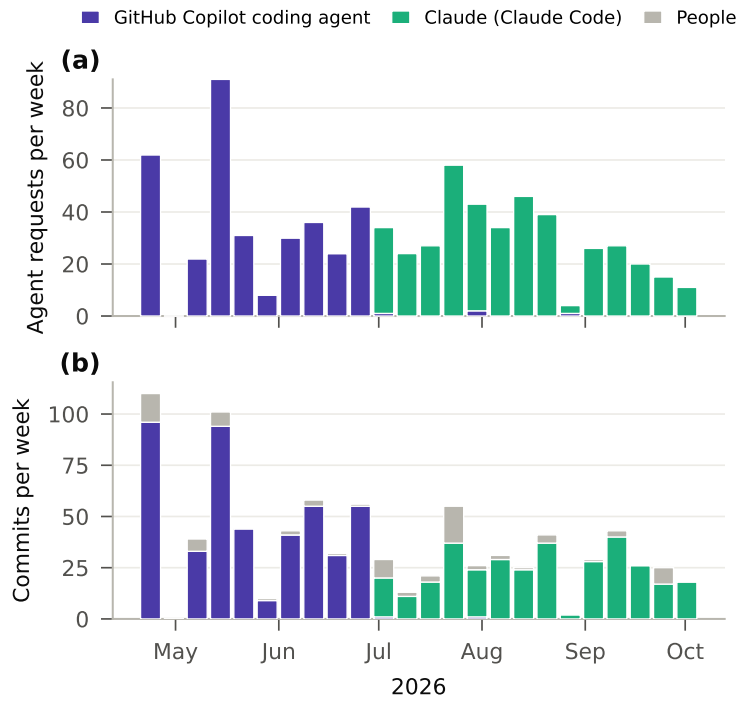


Figure S3: Generative-AI agent activity per week over the project. (a) Requests to an agent: comments and issue or pull-request descriptions written by a team member that address @copilot or @claude, plus pull requests the Copilot coding agent opened when assigned an issue. Each request starts one agent session. (b) Commits on all branches by author; commits that were later rebased onto another branch are counted once. Counts are from the public repository history (paper/figures/data/build_ai_usage.py).